Python Dictionary Methods

Important dictionary methods with simple examples and outputs

```
student = {
    "name": "Vishal",
    "age": 25,
    "city": "Delhi"
}
```

1. get()

Gets the value of a key. If the key does not exist, it returns None (or a specified default value).

Example:

```
print(student.get("name"))
print(student.get("marks", 0))
```

Output:

```
Vishal
0
```

2. keys()

Returns all keys of the dictionary.

Example:

```
print(student.keys())
```

Output:

```
dict_keys(['name', 'age', 'city'])
```

3. values()

Returns all values of the dictionary.

Example:

```
print(student.values())
```

Output:

```
dict_values(['Vishal', 25, 'Delhi'])
```

4. items()

Returns all key-value pairs as tuples.

Example:

```
print(student.items())

for key, value in student.items():
    print(key, value)
```

Output:

```
dict_items([('name', 'Vishal'), ('age', 25), ('city', 'Delhi')])

name Vishal
age 25
city Delhi
```

5. update()

Adds new key-value pairs or updates existing values.

Example:

```
student.update({"age": 26, "course": "MCA"})
```

```
print(student)
```

Output:

```
{'name': 'Vishal', 'age': 26, 'city': 'Delhi', 'course': 'MCA'}
```

6. pop()

Removes a specified key and returns its value.

Example:

```
age = student.pop("age")
print(age)
print(student)
```

Output:

```
25
{'name': 'Vishal', 'city': 'Delhi'}
```

7. popitem()

Removes and returns the last inserted key-value pair.

Example:

```
student = {
    "name": "Vishal",
    "age": 25,
    "city": "Delhi"
}
print(student.popitem())
```

Output:

```
('city', 'Delhi')
```

8. clear()

Removes all items from the dictionary.

Example:

```
student.clear()
print(student)
```

Output:

```
{}
```

9. copy()

Creates a copy of the dictionary.

Example:

```
student = {"name": "Vishal", "age": 25}
new_student = student.copy()
print(new_student)
```

Output:

```
{'name': 'Vishal', 'age': 25}
```

10. setdefault()

Returns the value of a key. If the key is missing, it creates the key with the default value.

Example:

```
student = {"name": "Vishal", "age": 25}
student.setdefault("city", "Delhi")
print(student)

student.setdefault("age", 30)
print(student["age"])
```

Output:

```
{'name': 'Vishal', 'age': 25, 'city': 'Delhi'}
25
```

11. fromkeys()

Creates a new dictionary from a sequence of keys, using the same value for each key.

Example:

```
keys = ["name", "age", "city"]

data = dict.fromkeys(keys, None)
print(data)

data = dict.fromkeys(keys, "Unknown")
print(data)
```

Output:

```
{'name': None, 'age': None, 'city': None}
{'name': 'Unknown', 'age': 'Unknown', 'city': 'Unknown'}
```

Quick Revision

Method	Purpose
get()	Get a value safely
keys()	Get all keys
values()	Get all values
items()	Get key-value pairs
update()	Add or update data
pop()	Remove a specific key
popitem()	Remove the last inserted pair
clear()	Remove everything
copy()	Create a copy
setdefault()	Get or create a key
fromkeys()	Create a dictionary from keys

Interview focus: get(), keys(), values(), items(), update(), pop(), and setdefault() are especially important to understand.